# LANDSLIDE INVENTORY AND MONITORING USING SENTINEL-1 SAR IMAGERY

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## ABSTRACT

An important application of differential SAR interferometry (DInSAR) and Persistent Scatterer Interferometry is landslide detection and monitoring. Several studies have been published, which make use of the entire spectrum of SAR data types available in the last 25 years. This paper describes a procedure to update landslide inventory maps using Sentinel-1 data. The paper briefly discusses the main advantages of the Sentinel-1 SAR data. Then it describes the data analysis procedure used to update landslide inventory maps using interferometric data and a number of additional information layers. The effectiveness of the procedure is illustrated by the results of a study area located in the Molise region, in Southern Italy.

## 1. INTRODUCTION

This paper describes a procedure to update landslide inventory maps using Sentinel-1 interferometric data. Landslide inventory maps contain information on landslide activity and are a basic input to landslide susceptibility and hazard analysis. Several works have been focused on the use of SAR (Synthetic Aperture Radar) interferometric data for landslide-related applications, e.g. see [1-9]. For a review of Persistent Scatterer interferometry, see [10].

New perspectives in landslide inventory and monitoring are now opened by the availability of the SAR data of the Sentinel-1 satellite of the European Space Agency. This type of SAR data offers very interesting characteristics. Firstly, Sentinel-1 data are acquired with a short revisiting cycle of 12 days, which will become of 6 days with the availability of Sentinel-1B satellite. This results in a reduced temporal decorrelation. Secondly, the Sentinel-1 data have a reduced orbital tube, which implies a reduced geometric decorrelation. Thirdly, the Sentinel-1 data acquired in the Interferometric Wide Swath mode, which is the standard acquisition mode for this sensor, cover wide areas. In addition, the image acquisitions are in background mode (images are acquired systematically according to a pre-defined observation scenario). A fourth and important aspect is that Sentinel-1 data are available free of charge to all data users. This represents an important advantage from the application point of view.

This paper describes the procedure adopted by the authors to update a landslide inventory using Sentinel-1 data. The effectiveness of the procedure is illustrated by using results obtained over the Molise region, in Southern Italy.

## 2. DATA ANALYSIS PROCEDURE

The procedure includes two main stages: a simplified Persistent Scatterer Interferometry analysis and a multi-layer GIS (Geographic Information System) analysis.

The first stage, which is described in [11, 12], is performed in the SAR geometry (azimuth and slant range geometry) using the available interferometric data. It is done both spatially and temporally with the aim of detecting areas affected by deformation. The main output of this analysis is a set of areas potentially affected by deformation.

The second stage (multilayer GIS analysis) consists in the integration of the interferometric-derived data with geological and geomorphological data to interpret and validate the detected deformation areas. This information can be used to update pre-existing landslide inventory maps.

The main steps of the two-stage procedure, see Figure 1, are briefly discussed below.

- The first step is interferogram generation. Given a stack of N complex SAR images, we generate the N-1 consecutive multi-look interferograms.
- The spatial analysis consists in a visual inspection of the wrapped interferograms to identify spatial patterns associated with potential deformation areas. This analysis only provides information on movements that are fast enough to be observed in short (typically 12 days) periods. Once the patterns are detected, the pairwise logic [13] is used to discard patterns due to other sources, e.g. topographic errors. The result of this step is a set of areas potentially affected by deformation.

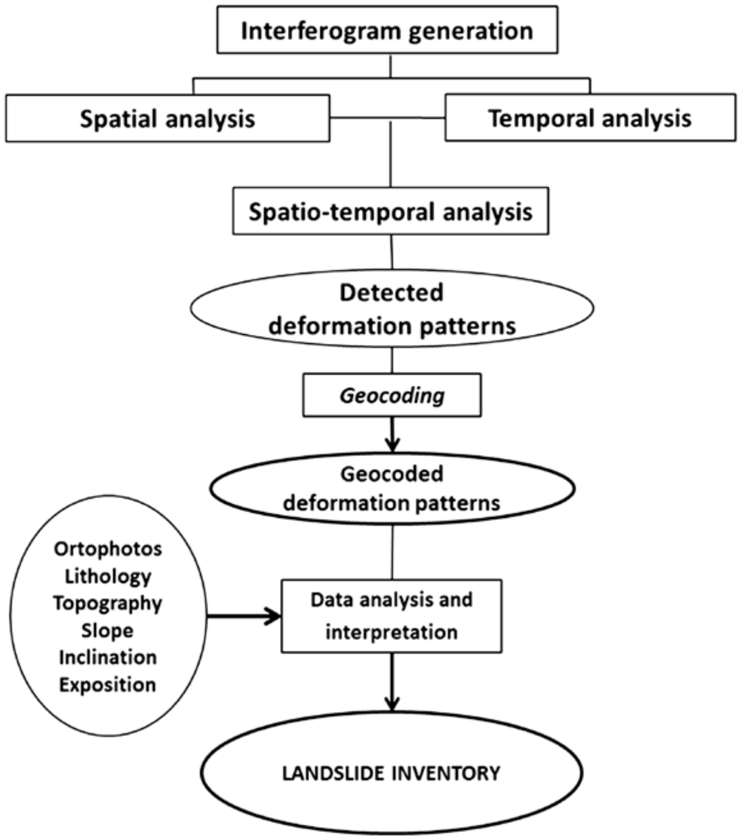


*Figure 1: Flow-chart of the procedure used in this study.*

- The third step is a temporal analysis, which consists in the generation of deformation time series over a set of coherent pixels. The procedure involves the phase unwrapping of the interferograms using the Minimum Coast Flow approach [14]. Only the pixels with coherence above a given threshold are unwrapped. The deformation time series are obtained by a direct integration of the unwrapped phase values and by transforming the phases into displacements.
- The fourth step involves the analysis of the map of accumulated deformation to search for additional spatial patterns characterized by slow deformation rates. The analysis of the time series is done with respect to a local stable reference to reduce the atmospheric effects.
- The fifth step is the spatio-temporal analysis. The potential areas of deformation that were detected in the previous steps are analysed together with the time series. This combined analysis is useful to detect phase unwrapping errors and to confirm or modify the shape of the detected deformation areas.
- The result of the previous step is the final set of detected deformation phenomena. This dataset is then geocoded in order to have it in the same reference system of the cartographic, geological and additional information.
- The last step is the multilayer GIS analysis. This involves a geological and geomorphological interpretation of the detected areas using different information layers in a GIS environment: a digital elevation model, slope, aspect, orthophotos, geo-lithological maps, existing landslide inventory maps, etc. At this step, the previously identified potential deformation areas are confirmed, discarded or modified and the results are used to update the existing landslide inventory maps.

## 3. RESULTS

The procedure described above was successfully used to study an area located in the Molise region, in Southern Italy. The study area is affected by a great number of landslide phenomena, see for details [15].

The analysis was focused on a single burst of a set of 14 Sentinel-1 Interferometric Wide Swath images (single polarization VV) acquired with ascending orbits. The images span a relatively short time period, from November 2014 to April 2015.

Figure 2 shows some examples of potential deformation patterns that were identified using a 12-day wrapped interferogram. Three main patterns are highlighted.

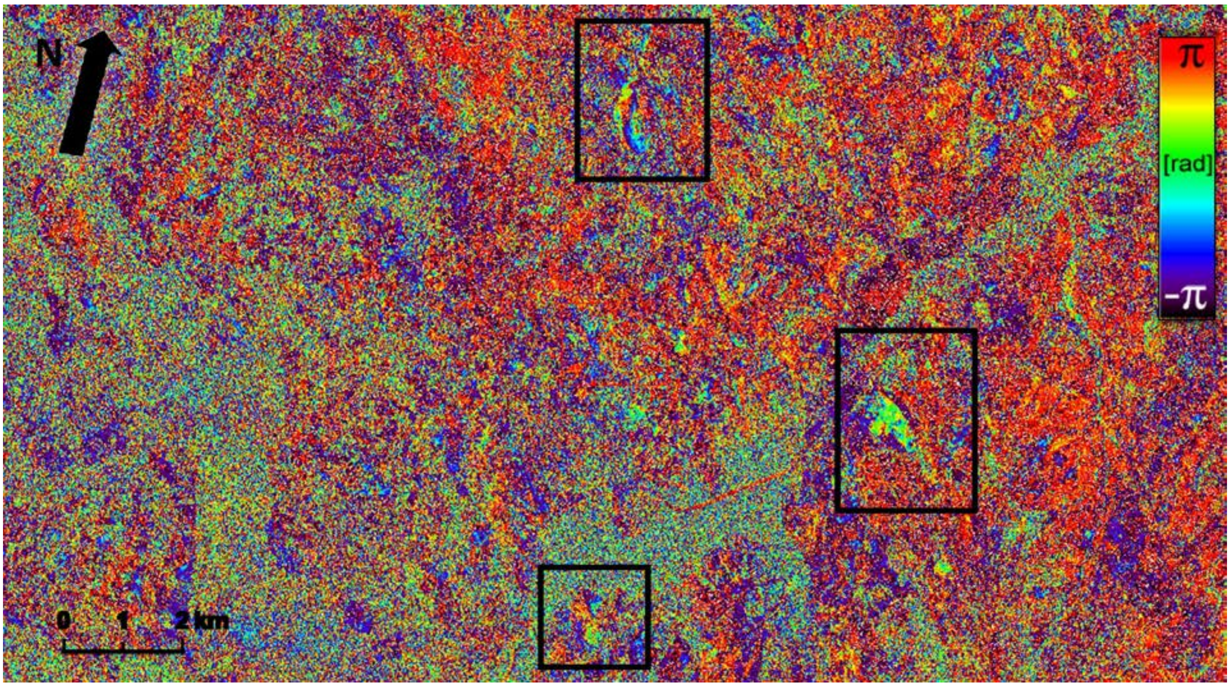


*Figure 2: Three potential deformation patterns identified in a 12-day wrapped interferogram.*

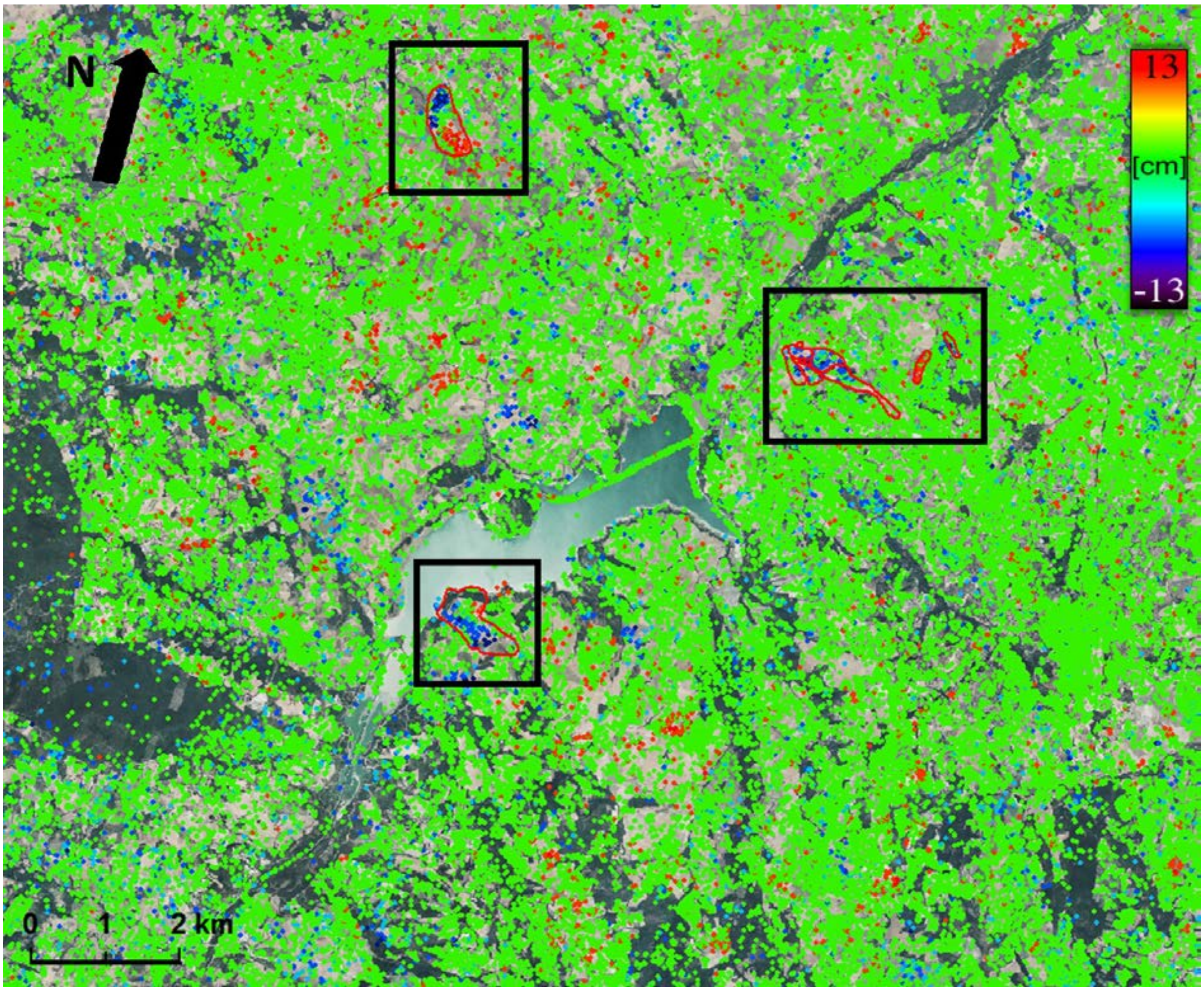


*Figure 3: Accumulated deformation map with some examples of landslides detected using the multilayer GIS analysis.*

Figure 3 shows an example of outcome of the multilayer GIS analysis: a set of confirmed landslides. The border of each landslide is shown in red. The landslides are superposed to the accumulated deformation map.

## 4. ACKNOWLEDGEMENTS

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